CALCULUS · UNIT 6: THE FUNDAMENTAL THEOREM OF CALCULUS

Coffee Shop Rush

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students manage a coffee shop's morning rush. The arrival rate and service rate define the line through an accumulation function, milk use is a rate integrated to find total use, and averages help plan staffing.

FTC Part 1 finds when the line is longest; FTC Part 2 computes totals.

Standards Addressed

- AP FUN-5 Fundamental Theorem of Calculus
- AP FUN-5.A Functions defined by integrals
- AP CHA-4 Accumulation and net change in context
- AP CHA-4.B Average value

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: "How long is the line at 8 a.m.?" Part 1 milk and customers.
Day 2	Part 2 accumulation functions.
Day 3	Part 3 staffing decision and plan.

Materials

- Student handout and Shop Data Cards
- Graphing calculators with integration
- Optional: a local coffee shop interview

How the Three Versions Differ

Level	What changes
CP2	Total change from rates, FTC Part 2 evaluations, average value, and a simple net-change plan.
CP1	Accumulation functions with FTC Part 1, peak line, chain rule with integrals, and a delivery model.
Honors	Extrema and inflection of integral functions, realistic line restarting at 0, staffing thresholds, and cost optimization.

Watch For (Common Misconceptions)

- Forgetting the initial amount in accumulation.
- Using the rate value instead of the integral for totals.
- Missing the chain-rule factor when the upper limit is not x.

Extension Ideas

- Collect arrival data at the school store.
- Simulate the line with a spreadsheet.
- Model a lunch rush at the cafeteria.

Answer Key – CP2

Part 1: Milk Supply

#	Task	Answer
1	Find $\int_0^4 (3 + 2t)\,dt$. What does it mean?	$[3t + t^2]_0^4 = 28$ gallons used from 6 to 10 a.m.
2	The shop has 12 gallons at 6 a.m. When does it run out? Solve $3t + t^2 = 12$.	$t^2 + 3t - 12 = 0 \rightarrow t \approx 2.27$ h (about 8:16 a.m.)
3	How many gallons are used from 7 to 9 a.m. (t = 1 to 3)?	$[3t + t^2]_1^3 = 18 - 4 = 14$ gallons
4	Find the average rate of milk use from 6 to 10 a.m.	$\frac{28}{4} = 7$ gallons per hour

Part 2: Customers

#	Task	Answer
5	Find A(0), A(2), and A(4). When is the rush the busiest?	60, 100, 60; at t = 2 (8 a.m.)
6	Find the total number of customers from 6 to 10 a.m.: $\int_0^4 A(t)\,dt$.	$240 + \frac{160}{\pi}(1 - \cos \pi) \approx 240 + 101.9 = 341.9 \approx 342$ customers
7	With 3 baristas (90 per hour), how many customers can be served in 4 hours?	360
8	Is 3 baristas enough for the morning? Explain.	Total capacity 360 > 342, but at the peak 100 > 90, so the line grows for a while.

Part 3: Plan

#	Task	Answer
9	Find the net change in the line from 6 to 10 a.m. with 3 baristas: $\int_0^4 (A(t) - 90)\,dt$.	$342 - 360 = -18$ (the line shrinks overall)
10	The line starts at 5 people. Use 5 + (your net change) to decide how many are in line at 10 a.m. (The line cannot be negative.)	$5 + (-18) < 0$, so the line is empty (0 people) by 10 a.m.
11	Write a staffing and milk plan for the manager.	Answers vary; e.g., 3 baristas and 30 gallons of milk.

Reflection

#	Prompt	Look for
R1	What does the integral of a rate tell you?	The total change in the quantity over the interval.
R2	Why is the average value of a rate useful to a manager?	It gives a single typical rate for planning.

Answer Key – CP1

Part 1: Line Length

#	Task	Answer
1	Find $L'(t)$ using FTC Part 1. When is the line growing?	$L'(t) = A(t) - 90 > 0$ when $\sin(\frac{\pi t}{4}) > 0.75 \rightarrow 1.08 < t < 2.92$
2	When is the line longest? Justify.	$t \approx 2.92$ h (about 8:55 a.m.), where L' changes from + to −
3	Find the maximum line length (technology allowed).	$L(2.92) = \int$ from 1.08 to 2.92 of $(A(x) - 90)\,dx \approx 12.2 \rightarrow$ about 12 people
4	Find $L(4)$ and interpret.	$L(4) \approx -3.0 < 0$, so the line empties shortly before 10 a.m. (about 9:54) and stays at 0.

Part 2: Milk & FTC

#	Task	Answer
5	Write the milk remaining as $G(t) = 12 - \int_0^t M(x)\,dx$ and find $G(2)$.	$G(2) = 12 - 10 = 2$ gallons
6	Find $\frac{d}{dt} \int_0^{(t^2)} (3 + 2x)\,dx$.	$(3 + 2t^2)(2t)$
7	A delivery adds 20 gallons at 8 a.m. Write the milk remaining for $t > 2$ and find when it runs out.	$G = 32 - 3t - t^2 \rightarrow t^2 + 3t - 32 = 0 \rightarrow t \approx 4.35$ h (after 10 a.m., so the delivery is enough)
8	Find the average value of $A(t)$ on $[0, 4]$.	$\frac{342}{4} \approx 85.5$ customers per hour

Part 3: Staffing

#	Task	Answer
9	With 2 baristas (60/hour), write $L_2(t) = 5 + \int_0^t (A - 60)\,dx$ and find $L_2(4)$.	$5 + 342 - 240 = 107$ people — far too many
10	With 3 baristas, show the line never exceeds 25.	The line peaks at about 12 people (at $t \approx 2.92$), which is under 25 ✓.
11	Write the staffing plan with the number of baristas by hour and the milk order.	Answers vary.

Reflection

#	Prompt	Look for
R1	How does FTC Part 1 let you find when the line is longest without integrating?	$L'(t) = A(t) - 90$, so you solve $A(t) = 90$ and test signs.

#	Prompt	Look for
R2	What assumptions in the model break down, and why?	The line cannot be negative; when it reaches 0 baristas wait for customers.

Answer Key – Honors

Part 1: Accumulation Analysis

#	Task	Answer
1	Find $g'(t)$ and $g''(t)$.	$g' = A(t) - 90$; $g'' = A'(t) = 10\pi \cos(\frac{\pi t}{4})$
2	Find the critical points of g and classify them.	$g' = 0$ at $t = \frac{4}{\pi} \sin^{-1}(0.75) \approx 1.08$ (min) and ≈ 2.92 (max)
3	Find the inflection point of g and interpret.	t = 2: arrivals are highest, so the line grows fastest.
4	Evaluate g(4) exactly.	$\int_0^4 (-30 + 40 \sin(\frac{\pi x}{4}))\, dx = -120 + \frac{320}{\pi} \approx -18.1$

Part 2: Realistic Line

#	Task	Answer
5	Starting with 5 people at 6 a.m. and 3 baristas, when does the line first empty? (Use $L(t) = 5 + g(t)$.)	Solve $5 + g(t) = 0 \rightarrow t \approx 0.18$ h (about 6:11 a.m.)
6	After the line empties it stays 0 until $A(t) > 90$ at $t \approx 1.08$. Find the line length at its peak ($t \approx 2.92$).	$\int$ from 1.08 to 2.92 of $(A - 90)\, dx \approx 12.2 \rightarrow$ about 12 people
7	Find the maximum number of baristas that would still let a line form at the peak.	3 (any staff serving < 100 per hour); with 4 (120/h) the line never forms after 6 a.m.
8	Find the average line length from 1.08 to 2.92 hours using technology.	About 6.1 people

Part 3: Optimization & Report

#	Task	Answer
9	Compare total cost for 3 baristas vs. 4 baristas from 6 to 10 a.m.	3 baristas: \$264 + lost sales ≈ 6.1 people × 110 min × \$0.50 ≈ \$335 → about \$599. 4 baristas: \$352 and almost no line (120 per hour > peak 100). Four baristas cost less overall.
10	Find $\frac{d}{dx} \int_x^{(3x)} (3 + 2u)\, du$.	$3(3 + 6x) - (3 + 2x) = 6 + 16x$
11	Write the staffing plan with a schedule by hour, justified with the FTC.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why does a function defined by an integral have its maximum where the integrand changes sign?	Its derivative is the integrand (FTC Part 1).

#	Prompt	Look for
R2	How did the real-world constraint (no negative line) change the model?	The accumulation had to restart at 0 once the line emptied.